

GenAI Powered Data Analytics

Summer Internship / Virtual Job Simulation

Tata iQ • Forage

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1. Program Overview

Self-paced Tata iQ GenAI Powered Data Analytics job simulation on Forage.

Simulated role: AI Transformation Consultant in a financial-services analytics team.

Business case: helping Geldium Finance understand and predict credit-card delinquency risk.

Core themes: EDA, data quality, predictive modeling, GenAI assistance and responsible AI.

2. Business Problem

Geldium Finance observed increasing credit-card delinquency and wanted a more structured, data-driven approach.

Existing process relied on historical trends and manual case handling.

Goal: identify at-risk customers earlier and support targeted intervention strategies.

Risk assessment should remain fair, explainable and responsible.

3. Task 1 — EDA & Risk Profiling

Understand dataset structure and key variables.

Identify missing, inconsistent or anomalous data before modeling.

Explore relationships between customer variables and delinquency.

Document early risk indicators and data-quality observations.

4. Data Quality & Missing Values

Assess why data is missing before choosing a treatment.

Possible approaches: deletion, statistical imputation, predictive imputation and carefully validated synthetic data.

GenAI can help detect gaps and suggest strategies, but final choices require analytical validation.

Reliable data is a prerequisite for reliable predictive modeling.

5. Risk Factors Discussed

Payment history and previous missed/late payments.

Credit utilization and potential financial pressure.

Debt-to-income ratio and ability to manage obligations.

Recent credit activity plus employment/income stability.

Risk indicators should be investigated rather than treated as automatic causal conclusions.

6. Synthetic Data

Synthetic data can supplement limited or incomplete datasets.

It should mimic relevant real-world patterns without exposing real customer information.

Validate generated data against real-world distributions and statistical expectations.

Use synthetic data as a supplementary tool, especially in regulated financial contexts.

7. Task 2 — Predicting Delinquency

Move from exploration to predictive modeling.

Consider approaches such as logistic regression, decision trees and neural networks.

Use GenAI to assist with model logic, code drafting and interpretation.

Validate model outputs before using them to support decisions.

8. Evaluation & Responsible AI

Use appropriate hold-out or cross-validation methods.

Assess classification performance beyond simple accuracy.

Check calibration, error patterns and subgroup performance where relevant.

Prioritize explainability, fairness, transparency and human oversight.

9. Skills Gained

Exploratory Data Analysis and data-quality assessment.

Predictive analytics, model selection and model-validation concepts.

GenAI-assisted analysis and prompt-based exploration.

Business communication, strategic thinking and responsible AI reasoning.

10. Conclusion

The simulation provided an end-to-end view of a data/AI project in a financial-services setting.

Key learning: good modeling starts with reliable data and clear problem framing.

Future extension: implement the workflow in Python, perform quantitative EDA, train and compare models, and build a monitoring dashboard.

11. Evidence of Internship Activity

Selected screenshots from the supplied Forage learning path

The screenshot shows the Forage learning path interface. The top navigation bar includes 'Dashboard', 'Job Simulations', and 'Jobs'. The main content area is titled 'Intro & Scenario' and features a sidebar with a progress indicator showing three steps: 'Intro & Scenario', 'Exploratory data analysis and risk profiling', and 'Predicting delinquency with AI'. The main content area includes a 'Your Role' section with a briefcase icon and a 'Your Goal' section with a target icon. The 'Your Role' section lists three bullet points: 'You are an AI transformation consultant at Tata iQ, specializing in data-driven decision-making for financial services.', 'You work within a larger data analytics team, collaborating with business leaders to develop AI-powered solutions.', and 'Your team has been tasked with helping Geldium Finance, a financial services company, reduce its high credit card delinquency rate by performing advanced analytics and building AI/ML models with the assistance of GenAI.' The 'Your Goal' section lists three bullet points: 'Your primary objective is to analyze customer data and predict delinquency risks using AI-driven techniques.', 'You will develop a recommendation framework to help the Head of Collections at Geldium determine the best intervention strategies for at-risk customers.', and 'Throughout the project, you will ensure that AI-driven solutions are ethical, explainable, and effective in supporting responsible financial decision-making.'

The screenshot shows the TATA GenAI Powered Data Analytics task overview page. The top navigation bar includes 'Dashboard', 'Job Simulations', and 'Jobs'. The main content area is titled 'Task overview' and features a sidebar with a progress indicator showing three steps: 'Intro & Scenario', 'Exploratory data analysis and risk profiling', and 'Predicting delinquency with AI'. The main content area includes a 'What you'll learn' section with a book icon and a 'What you'll do' section with a magnifying glass icon. The 'What you'll learn' section lists three bullet points: 'How to conduct exploratory data analysis (EDA) using GenAI.', 'Techniques to handle missing values and ensure data quality.', and 'Understanding customer risk factors for delinquency.' The 'What you'll do' section lists three bullet points: 'Identify key datasets required to predict delinquency.', 'Perform an exploratory analysis on provided datasets, using GenAI tools to assist with summarization, treatment of missing data, risk profiling, and synthetic data creation.', and 'Document your findings, including data patterns and anomalies that may impact predictions.'

The screenshot shows the TATA GenAI Powered Data Analytics task overview page, focusing on the 'What you'll learn' and 'What you'll do' sections. The 'What you'll learn' section lists three bullet points: 'How to leverage GenAI for predictive modeling without coding.', 'Common techniques for credit risk modeling (e.g., decision trees, logistic regression, neural networks).', and 'Evaluating model performance and ethical considerations (bias, explainability, fairness).'. The 'What you'll do' section lists three bullet points: 'Use GenAI tools to generate model logic for predicting customer delinquency.', 'Choose and justify the best predictive approach.', and 'Outline a plan to evaluate model accuracy and reliability.'

Thank You

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